WEIMIN WU

Northwestern University • wwm@u.northwestern.edu • (+1) 773-681-4449
Homepage • Google Scholar

EDUCATION

McCormick School of Engineering, Northwestern University, Evanston, USA Sep 2023 – Present
Ph.D. in Computer Science & Biochemistry and Molecular Genetics, Co-advised by **Han Liu** and **Yaping Liu**
McCormick School of Engineering, Northwestern University, Evanston, USA Sep 2023 – Jun 2026
M.S. in Computer Science, Advised by **Han Liu**
School of Mathematical Sciences, Fudan University, Shanghai, China Sep 2019 – Jun 2023
B.S. in Computational Mathematics, Outstanding Graduate of Fudan University, Advised by **Tao Chen**

RESEARCH INTERESTS

My research focuses on **recursively self-improving AI agents for scientific discovery**, advancing noninvasive early cancer detection using cell-free DNA and personalized cancer treatment through DNA/RNA vaccine design. My work spans four interconnected areas: **(i) genomic foundation models**, including GenomeOcean for genomic sequence understanding and design; **(ii) model infrastructure**, including Genome-Factory for genomic model deployment and adaptation; **(iii) sequence design**, including D-DOIT for DNA/RNA design toward personalized cancer vaccines; and **(iv) recursively self-improving scientific agents**, including SciMARS, toward integrating models, tools, and feedback for iterative scientific discovery.

GRANTS & AWARDS

NSF's NAIRR grant (NAIRR240377, **$460,000**): Towards the world's leading and largest genomic foundation model.

2026 Ph.D. Student Research Award, Computer Science, Northwestern University

INTERNSHIP EXPERIENCES

NVIDIA, CA, USA | Genomics Deep Learning Engineering Intern. Jun. 2026 – Sep. 2026
- Advisor: **Pankaj Vats**, **Zhihan Zhou** & **Tong Zhu**
- Develop genomic foundation models for genome annotation.

Quest Diagnostics, NJ, USA | Machine Learning Researcher (Collaborate with NVIDIA) Jun. 2025 – Sep. 2025
- Advisor: **Paz Polak**; Collaborator: **Pankaj Vats**
- Improve the existing method from 0.8 to 0.91 based on the genomic foundation model.

JOURNAL PUBLICATIONS

***DNABERT-S: Learning Species-Aware DNA Embedding with Genome Foundation Models*, Bioinformatics, 2025**
Zhihan Zhou*, **Weimin Wu***, Zhong Wang, Han Liu et al. **(*: joint first author)**
Contrastive representation learning for species-aware DNA embeddings and genomic sequence analysis.

***Uncovering the Genomic Manifold via Scalable Learning from the Global Microbiome*, 2025**
Weimin Wu*, Zhihan Zhou*, Yaping Liu, Han Liu, Zhong Wang et al. *(Under Review of **Nature Methods**)*
A 4B-parameter generative foundation model pretrained on 64 A100 GPUs for genomic sequence modeling.

CONFERENCE PUBLICATIONS

***D-DOIT: Training-free Adaptation of Discrete Diffusion via Doob's h-Transform*, NeurIPS, 2026**
Jieke Wu*, Qijie Zhu*, **Weimin Wu***, Zeqi Ye, Minshuo Chen, Han Liu **(*: joint first author)**
Training-free guidance for discrete diffusion via Doob's h-transform, applied to DNA and protein design.

***Genome-Factory: An Integrated Library for Tuning, Deploying, and Interpreting Genomic Models*, ICML, 2026**
Weimin Wu, Zhong Wang, Han Liu et al.
A unified software framework for model adaptation, deployment, and interpretation, applied to genomics.

***Sci2Pol: Evaluating and Fine-tuning LLMs on Scientific-to-Policy Brief Generation*, ICLR, 2026**
Weimin Wu, Dashun Wang, Han Liu et al.
The first benchmark and training dataset to evaluate and fine-tune LLMs on scientific-to-policy brief generation.

***In-Context Deep Learning via Transformer Models*, ICML, 2025**
Weimin Wu, Han Liu et al.
Prove the transformer's capability to simulate the training process of deep models via in-context learning.

***In-Context Learning as Conditioned Associative Memory Retrieval*, ICML, 2025**
Weimin Wu, Han Liu et al.
The first interpretation of in-context learning from the perspective of dense associative memory models (Hopfield).

***On Statistical Rates of Conditional Diffusion Transformers: Approximation and Estimation*, ICLR, 2025**
Jerry Hu*, **Weimin Wu***, Minshuo Chen, Han Liu et al.
The first comprehensive statistical theory (approximation and estimation) for conditional diffusion transformers.

***On Statistical Rates and Provably Efficient Criteria of Latent Diffusion Transformers (DiTs)*, NeurIPS, 2024**
Jerry Hu*, **Weimin Wu***, Han Liu
The first statistical (approximation and estimation) and computational theory for diffusion transformers.

***Universal Approximation with Softmax Attention*, ICML, 2026**
Jerry Hu**, Weimin Wu**, Han Liu et al.
Prove that the attention layer is a universal approximator for sequence-to-sequence functions.

***On Structured State-Space Duality*, ICML, 2026**
Jerry Hu**, Weimin Wu**, Han Liu et al.
Prove the equivalence between a structured state-space model and a masked attention mechanism.

***Learning Manifold Data with Flow Matching*, ICML, 2026**
Sophia Pi**, Weimin Wu**, Han Liu et al.
A comprehensive statistical theory study of flow-matching transformers with low-dimensional manifold.

***POLO: Preference-Guided Multi-Turn Reinforcement Learning for Lead Optimization*, KDD, 2026**
Ziqing Wang, **Weimin Wu**, Han Liu, Kaize Ding et al.
Sample-efficient multi-turn reinforcement learning with preference feedback, applied to molecular lead optimization.

PREPRINTS

***SciMARS: Multi-Agent Recursive Self-Improvement for Closed-Loop Scientific Discovery*, arXiv, 2026**
Weimin Wu, Han Liu
A recursive self-improvement framework for multi-agent system in closed-loop scientific discovery.

***DiMOS: Doob-Guided Inference-Time Multi-Objective Search for Scientific Design*, arXiv, 2026**
Ziqing Wang*, Qijie Zhu*, **Weimin Wu***, Zeqi Ye, Minshuo Chen, Han Liu, Kaize Ding **(*: joint first author)**
Training-free multi-objective search with discrete diffusion, applied to DNA, RNA, and protein design.

***Cell-JEPA: Latent Representation Learning for Single-Cell Transcriptomics*, arXiv, 2026**
Ali ElSheikh*, Rui-Xi Wang*, **Weimin Wu***, Aly A. Khan, Han Liu et al.
Latent representation learning with a joint-embedding predictive architecture, applied to single-cell transcriptomics.

***Discrete Flow Matching Policy Optimization*, arXiv, 2026**
Maojiang Su, **Weimin Wu**, Han Liu et al.
A unified reinforcement learning framework for policy optimization of discrete flow-matching models.

***Instance-aware Model Ensemble With Distillation For Unsupervised Domain Adaptation*, arXiv, 2022**
Weimin Wu, Tao Chen et al.
An adaptive ensemble method for unsupervised domain adaptation in computer vision.

INVITED TALKS

Feinberg School of Medicine, Northwestern University 02/20/2026
- Advancing Science with World's Leading Genomic Foundation Models.

School of Medicine, Shanghai Jiao Tong University 12/22/2025
- Advancing Science with World's Leading Genomic Foundation Models

International Society for Computational Biology, Oral 07/23/2025
- DNABERT-S: Learning Species-aware DNA Embedding with Genome Foundation Models.

Harvard Medical School, Harvard University 03/14/2024
- DNABERT-S: Learning Species-aware DNA Embedding with Genome Foundation Models.

ENTREPRENEURSHIP

Founder and President of Quantitative Investment Association (QIA), China Jun. 2021 – Jun. 2023
- The largest non-profit organization in quantitative finance in China.

TEACHING EXPERIENCE

CS110: Introduction to Computer Programming, Northwestern University Fall 2024
- Conducted weekly discussion sections and office hours for 100+ undergraduate students.
- Graded assignments and provided constructive feedback on Python programming projects.

PROFESSIONAL SERVICE

Reviewer of *ICLR 2026, NeurIPS 2025, ICML 2025, ICLR 2025*

PATENTS

***Instance-aware Model Ensemble With Distillation For Unsupervised Domain Adaptation*, 2023**
Weimin Wu, Tao Chen et al., Chinese Patent, No. 202310743718.9

SUPERVISED RESEARCH ASSISTANTS

Maojiang Su (University of Science and Technology of China) 2025
- Next Position: CS Ph.D. at Northwestern University, First author paper @ ICML 2025

Morris Huang (National Taiwan University) 2024
- Next Position: CS Ph.D. at the University of North Carolina at Chapel Hill, First author paper @ ICLR 2025

Sophia Pi (Northwestern University) 2024
- Next Position: CS Ph.D. at University of Pennsylvania, One paper @ NeurIPS 2024